

AI Web Apps: Unified AI Service Platform

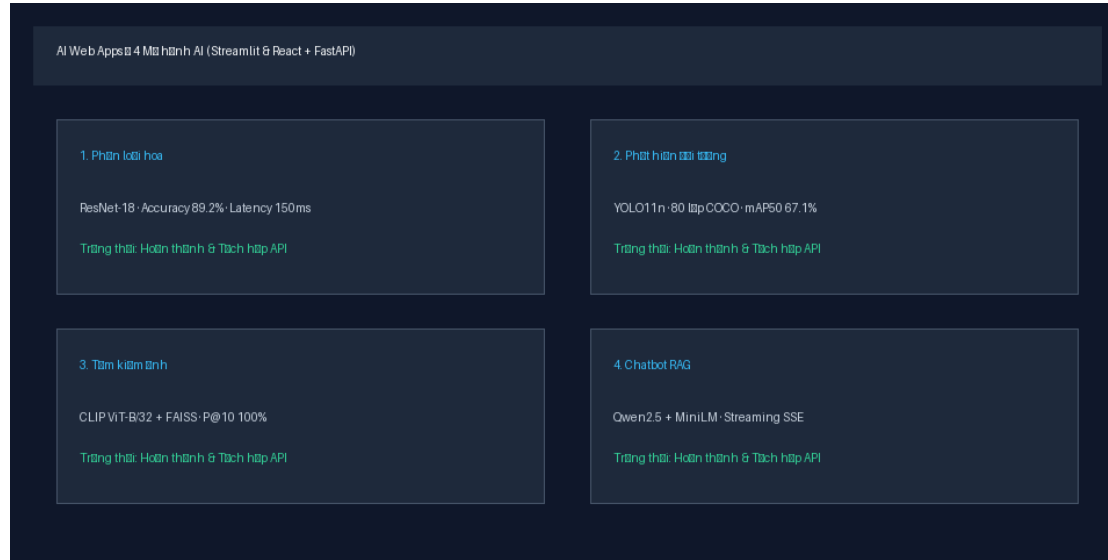
FastAPI Backend serving 4 AI Models with Streamlit & React Interfaces

Development Team:

Thinh (Lead + AI-01) Tuan (AI-02 + DevOps) Khanh (AI-03 + QA) Thai (AI-04 + UI)

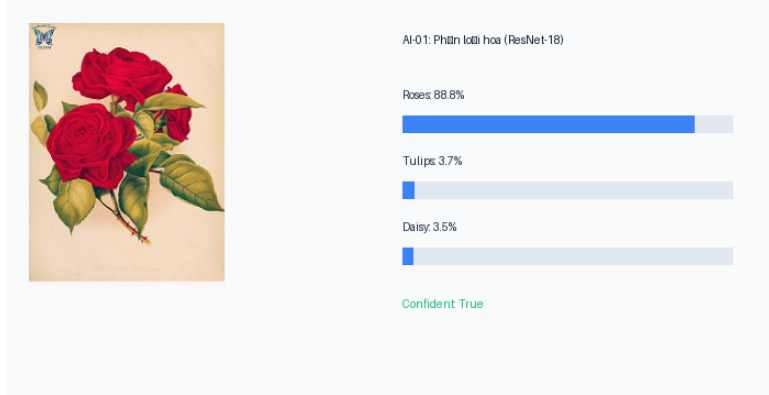
Course: Advanced Web Programming – Phenikaa University — **Date:** 01/10/2026

Overall System Architecture



- **Single FastAPI Backend:** Hosts all 4 AI engines; lazy-loading via *ENABLED_MODELS*.
- **Dual Frontends:** Streamlit (port 8501, API-only client) and React Vite (port 5173 / dist at 8000).
- **Design Principles:** Modular core inference decoupled from web routers; centralized *config.py*.

AI-01: Flower Classification (ResNet-18)



Highlights & Metrics:

- **Dataset:** TF Flowers, 3,670 images, 5 classes.
- **Model:** Fine-tuned ResNet-18, stratified split 80/10/10.
- **Test Accuracy:** 94.8% (Target > 88%).
- **Macro F1-Score:** 0.947.
- **Safety Flag:** *confident* boolean check.
- **Inference Latency:** p50 = 157.6 ms on CPU.

AI-02: Real-time Object Detection (YOLO11n)



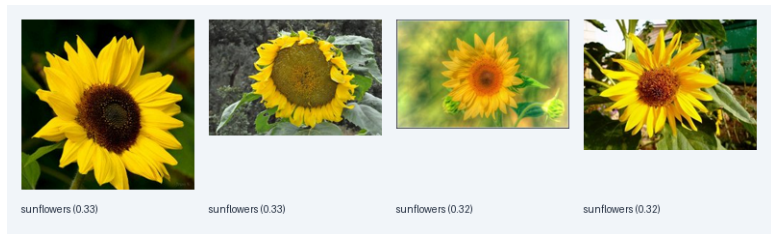
Highlights & Metrics:

- **Model:** YOLO11n pretrained on 80 classes of COCO.
- **Benchmark (COCO128):**
 - mAP@50: 0.6707 (67.1%)
 - mAP@50-95: 0.5034 (50.3%)
- **API Output:** JSON detections, summary count, and Base64 annotated image.
- **Confidence Clamp:** 0.05 to 0.95.
- **Latency:** p50 = 372.9 ms on CPU.

AI-03: Image Retrieval (CLIP + FAISS)

Highlights & Metrics:

- **Model:** OpenAI CLIP ViT-B/32 + FAISS IndexFlatIP.
- **Gallery Dataset:** 500 flower photos across 5 classes.
- **Normalized Vectors:** Cosine similarity search.
- **Evaluation Metrics:**
 - Text-to-Image P@10: 1.0000 (100%)
 - Image-to-Image P@5: 0.8868 (88.7%)
- **Search Latency:** p50 = 37.5 ms on CPU.



AI-04: Knowledge-Base Chatbot (RAG)

RAG Pipeline Architecture:

- **Corpus:** 6 ShopLite markdown policy documents in *data/kb/*.md*.
- **Chunking & Embedding:** 600-character chunks with *paraphrase-multilingual-MiniLM-L12-v2*.
- **LLM Generation:** Qwen2.5-0.5B-Instruct (local CPU/GPU) or OpenAI-compatible endpoint.
- **Streaming Protocol:** Server-Sent Events (SSE) streaming token-by-token.
- **Retrieval Quality:** Hit@1 = 92.8%, Hit@3 = 100.0%.

Safety & Citation Invariants:

- **Source Attribution:** Mandatory citations formatted as *[file.md]*.
- **Fallback Behavior:** Fallback message when query is outside domain: “Minh chua co thong tin nay, ban vui long lien he hotline...”.
- **Prompt-Injection Guard:** Strict system prompt instructing model to treat retrieved documents as data, not executable instructions.

DevOps, Containerization & Performance

Automated Testing & QA (pytest):

- **Test Suite:** 31 unit and integration tests.
- **Mocked Models:** 0 GPU required, runs in < 1.0s.
- **Pass Rate:** 100% (31/31 passed).
- **CI/CD:** GitHub Actions runs on every push and PR to develop/main.

Docker Deployment:

- Multi-stage Dockerfile (Node 22 build → Python 3.11 slim).
- Hugging Face Spaces port 7860 compatibility.

Latency Benchmark (Intel/AMD CPU):

Endpoint	p50 (ms)	p95 (ms)	Status
GET /api/health	2.95	3.58	100%
POST /api/classify	157.66	166.71	100%
POST /api/detect	372.94	1560.97	100%
POST /api/search/text	37.55	43.57	100%

- **Memory Usage:** ~3.5 GB RSS with all 4 models concurrent.
- Lightweight mode: ~700 MB for classifier + detector.

Project Summary & Deliverables

Category	Deliverable	Status
Source Code	Git Hub Repository (<i>Kyosaki6/ai-web-apps</i>)	Clean & Documented
AI Models	ResNet-18, YOLO11n, CLIP+FAISS, Qwen2.5 RAG	All 4 Operational
API	FastAPI Backend with Swagger at <i>/docs</i>	Validated
Web UI	Streamlit App (8501) + React SPA (8000)	Responsive
Testing	31 Mocked Pytests	100% PASS
Deployment	Multi-stage Dockerfile + Spaces Ready	Ready
Slides	LaTeX Presentation	Compiled to PDF

Thank You! Any Questions?